

# Absa Bank Zambia — Frontend Functional Requirements v2.0

Customer Lifecycle Prediction System · Enterprise Phase · August 6, 2026

**Supersedes:** FRONTEND-REQUIREMENTS.md (PoC v1.0, July 20, 2026) **Status:** Draft for Frontend Dev Team Review **Backend Reference:** Decision Intelligence Service v2.0 TDD + Database Design Spec v2.1

## 1. What's New in v2.0

| #  | Change                                                                                     | Rationale                                                                 |
|----|--------------------------------------------------------------------------------------------|---------------------------------------------------------------------------|
| 1  | <b>Branch Manager Dashboard</b> — promoted from deferred                                   | Layer 1-3 services now live; branch-level aggregation ready               |
| 2  | <b>Data Scientist: Full Model Ops</b> — champion/challenger, feature drift, SHAP waterfall | Prediction Service operational; model registry exists                     |
| 3  | <b>NBA with LLM Explanations</b> — human-readable SHAP-derived reasoning                   | Decision Intelligence Service v1.0 LLM pipeline live                      |
| 4  | <b>Campaign Management</b> — full campaign lifecycle UI                                    | <b>campaigns</b> schema exists in DB v2.1; Campaign Engine operational    |
| 5  | <b>Customer Journey Visualizer</b> — multi-touchpoint journey mapping                      | v2.0 target: "Customer journey optimization across multiple interactions" |
| 6  | <b>Feedback Loop / Action Outcomes</b> — measure RM actions → retention lift               | Closed-loop system required for RL (v1.1 roadmap)                         |
| 7  | <b>Priority Queue</b> — RM daily action queue ranked by Priority Score                     | Priority Engine operational; replaces ad-hoc table sorting                |
| 8  | <b>PDF Export</b> — one-page RM meeting summary (promoted from deferred)                   | Operations requirement                                                    |
| 9  | <b>WebSocket Notifications</b> — real-time alerts via WS                                   | System maturity; Redis pub/sub backend ready                              |
| 10 | <b>Full Audit Trail Viewer</b> — searchable, filterable, exportable                        | Regulatory compliance for Bank of Zambia                                  |

## 2. Architecture & Technology Constraints (Unchanged)

| Constraint       | Requirement                                                           |
|------------------|-----------------------------------------------------------------------|
| Framework        | Vue 3 (Composition API) + TypeScript                                  |
| Build Tool       | Vite                                                                  |
| State Management | Pinia                                                                 |
| Router           | Vue Router 4                                                          |
| UI Components    | Custom components or vendored PrimeVue (no npm CDN — bank air-gapped) |

| Constraint     | Requirement                                                                  |
|----------------|------------------------------------------------------------------------------|
| Charts         | Chart.js via vue-chartjs (vendored)                                          |
| Deployment     | Internal bank network — no internet, no CDN, on-premise Ubuntu server, Nginx |
| Backend API    | FastAPI Gateway (port 8080) — consumes 3-layer AI services                   |
| Authentication | Internal LDAP/Active Directory (no social logins)                            |
| Browser        | Chromium-based only (bank-standard locked-down desktops)                     |
| Screen         | 1920×1080 minimum — desktop only                                             |

### 3. User Roles & Access Control (v2.0 Updated)

#### 3.1 Role Matrix

| Role                 | v2.0 Permissions                                                                                                                           |
|----------------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| Relationship Manager | View own portfolio, drill into customers, see NBA + LLM explanations, log actions, mark outcomes, view daily priority queue                |
| Branch Manager       | View all branch customers, portfolio analytics, churn risk heatmap, team performance, campaign performance, churn forecast                 |
| Data Scientist       | Model performance dashboards, champion/challenger A/B, SHAP waterfall, feature drift, prediction logs, model registry, retraining triggers |
| Operations           | ETL pipeline status, data quality scores, audit logs, system health, service status dashboard                                              |
| Admin                | User provisioning, role assignment, API key management, session management, full audit log access                                          |

#### 3.2 Authentication Flow (Unchanged)

- Login via LDAP username/password
- Session timeout: 30 minutes inactivity (configurable)
- Role determined by LDAP group membership
- No self-registration — accounts provisioned by IT
- JWT + refresh token rotation (15-min access, 7-day refresh)

### 4. Screens — Relationship Manager (v2.0 Enhanced)

#### 4.1 RM Dashboard (Landing Page)

##### FR-V2-DASH-01: Summary Cards (Enhanced)

- 5 KPI cards: Total Customers, At Risk count + %, Dormant count + %, Actions Due Today, **Actions Completed This Week** (new)
- Each card clickable — filters the customer table below
- At Risk % colour-coded: green <5%, amber 5-15%, red >15%

##### FR-V2-DASH-02: Priority Queue (NEW)

- **Replaces** simple customer table sorting
- Ranked list of top 20 customers by Decision Intelligence Priority Score (0-100)
- Each row shows: Priority Rank, Customer Name, State, Health Score, Churn %, NBA Recommendation, Priority Score bar, **Estimated Revenue Impact (ZMW)** (new)
- Auto-refreshed after nightly batch; manual refresh available
- Priority formula displayed as tooltip: Business Value × Urgency × Acceptance Probability × Retention Impact

#### FR-V2-DASH-03: Priority Alerts (Enhanced)

- Top 5 alerts ranked by: Priority Score drop > 15 pts, Health Score drop > 20 pts, state transition to At Risk/Dormant/Churned, significant CLV change > 25%
- Each alert links to customer detail
- Alerts persist until acknowledged (timestamp + RM ID)
- **NEW: Alert severity badges** — Critical (red), High (amber), Info (blue)

#### FR-V2-DASH-04: Customer Table (Enhanced)

- Columns: Priority Rank, Customer Name, State (icon + label), Health Score (gauge), Churn % (colored), CLV (ZMW), NBA (truncated), **Last Contact Date** (new), **Days Since Last Action** (new)
- Default sort: Priority Score descending
- Search: name, account number, customer ID, **NBA type** (new)
- Filter: state, branch, product type, **priority tier** (new: Critical/High/Medium/Low)
- Pagination: 25 rows/page
- Click row → Customer Detail

#### FR-V2-DASH-05: Period Selector

- Dropdown: Current Snapshot vs historical dates (back to earliest feature snapshot)
- Default: Latest

### 4.2 Customer Detail Screen (v2.0 Enhanced)

#### FR-V2-CUST-01: Customer Profile Card

- Customer name, account number, branch, segment, tenure, assigned RM
- **NEW: Customer Journey Stage** — lifecycle stage indicator (Onboarding / Engaged / Mature / Declining / At Risk)
- Link to core banking system (external)

#### FR-V2-CUST-02: AI Health Score Gauge

- Circular gauge: Health Score (0-100)
- Colour: green 70+, amber 40-69, red < 40
- Component scores: Churn Risk, CLV Percentile, Behavioural Score
- Trend indicator: up/down arrow + Δ since last period (30 days)

#### FR-V2-CUST-03: State Timeline (Enhanced)

- Horizontal timeline: state changes over time
- States: Active (green), At Risk (amber), Dormant (grey), Churned (red)
- Hover: transition reason, trigger features, date
- **NEW: Markov next-state prediction overlay** — dotted line showing predicted next state with probability

**FR-V2-CUST-04: SHAP Waterfall (PROMOTED from PoC deferred)**

- Full SHAP waterfall chart showing feature contributions to current churn prediction
- Red bars: features pushing toward churn; Blue bars: features pushing away from churn
- Base value + cumulative contributions = final prediction
- **NEW: LLM-generated explanation** — natural language sentence below chart (e.g., "This customer's churn risk is elevated primarily due to 45 days of inactivity and a 32% drop in average transaction value, partially offset by strong product holdings.")

**FR-V2-CUST-05: NBA Recommendations (Enhanced)**

- Ranked list: 3-5 Next Best Actions
- Each shows: priority rank, action description, **LLM explanation** (1-2 sentences), predicted impact (High/Medium/Low), confidence %, estimated revenue impact (ZMW), estimated churn reduction (%), **recommended channel** (new: RM Call / Branch Visit / Digital / SMS), associated campaign name
- **NEW: Alternative Actions** — expandable section showing 2-3 alternative actions ranked below the top recommendation
- **[Log Action]** button opens form: action type dropdown, channel, notes, outcome dropdown (Contacted / Not Reachable / Declined / Accepted / Pending)
- Logged actions stored with: RM ID, customer ID, action taken, channel, timestamp, NBA recommendation ID, campaign ID

**FR-V2-CUST-06: Action History (Enhanced)**

- Reverse-chronological action log
- Filterable by: action type, date range, **outcome** (new)
- Columns: date, action, channel, outcome (colored badge), RM name, NBA reference
- **NEW: Outcome trend** — small inline chart showing action outcomes over time (Accepted/Declined/Pending ratio)

**FR-V2-CUST-07: Customer Journey Map (NEW)**

- Visual Sankey or flow diagram of customer's product holdings and interaction touchpoints
- Nodes: Products held (Savings, Current, Loan, Card, Insurance, etc.)
- Edges: Interaction channels used (Branch, Digital, ATM, RM Call)
- Timeline scrubber to view journey evolution over months

**5. Screens — Branch Manager (PROMOTED from Deferred)****5.1 Branch Manager Dashboard****FR-V2-BM-01: Branch KPIs**

- Aggregate KPI cards: Total Customers, At Risk %, Dormant %, Churn Rate (monthly), **Avg Portfolio Health Score** (new), **Actions Completion Rate** (new)
- $\Delta$  indicators vs previous month (green up / red down)
- Peer branch comparison (anonymised ranking: "Your branch: 3rd of 12")

**FR-V2-BM-02: Team Performance Table**

- Columns: RM Name, Customers Assigned, At Risk %, Actions Logged (month), **Actions Completed** (new), **Completion Rate %** (new), Avg Health Score, **Avg Days to Contact** (new)
- Sort by any column
- Click RM → filtered view of their portfolio
- **NEW: Performance trend sparkline** per RM (last 6 weeks)

#### FR-V2-BM-03: Churn Forecast (Enhanced)

- Projected churn count: next 30/60/90 days
- Breakdown by segment (Premium, Mass Market, SME)
- **NEW: Confidence bands** on forecast chart
- **NEW: "What-if" slider** — simulate impact of 10%/25%/50% intervention rate on projected churn

#### FR-V2-BM-04: Churn Risk Heatmap (Enhanced)

- Grid: rows = top 50 at-risk customers, columns = weeks
- Cell colour: churn probability gradient (green→red)
- Click cell → customer detail
- **NEW: Segment filter** — Premium / Mass Market / SME tabs

#### FR-V2-BM-05: Campaign Performance (NEW)

- Active campaigns table: Campaign Name, Target Segment, Customers Targeted, Responses, Conversion Rate, Revenue Generated
- Campaign detail drill-down: response timeline, segment breakdown, top-performing RMs
- Compare campaigns side-by-side

---

## 6. Screens — Data Scientist (PROMOTED + Enhanced)

### 6.1 Model Operations Dashboard

#### FR-V2-DS-01: Model Metrics (Enhanced)

- AUC-ROC, Precision, Recall, F1, Log Loss for champion model
- Time-series chart: 30/60/90 day trends
- Threshold alert lines (configurable per metric)
- **NEW: Segment-level metrics** — performance broken down by customer segment

#### FR-V2-DS-02: Champion vs Challenger (PROMOTED)

- Side-by-side comparison cards: Model Name, Version, Traffic Split %, Win Rate, Statistical Significance (p-value)
- Metric comparison table: AUC-ROC, Precision, Recall, F1 — champion vs challenger
- **[Promote Challenger]** button with confirmation dialog
- Traffic split slider (0-100%) for A/B test configuration

#### FR-V2-DS-03: Feature Drift Monitor (Enhanced)

- Table: Feature Name, Training Distribution (histogram thumbnail), Current Distribution (histogram thumbnail), PSI Score, Drift Status (OK/Warning/Critical)
- Threshold: PSI >0.1 warning, >0.25 critical (configurable)
- **NEW: Drift timeline** — per-feature PSI trend over last 30 days

- **NEW: Automated alert** — email/Slack when drift exceeds threshold (configurable)

#### FR-V2-DS-04: Prediction Log Browser (Enhanced)

- Searchable table: Timestamp, Customer ID, Prediction Type (Churn/CLV/Health), Probability, SHAP Top 3 Features, Model Version
- Filter by: date range, prediction type, model version, **segment** (new)
- Export to CSV
- **NEW: Click row → full SHAP waterfall + LLM explanation for that prediction**

#### FR-V2-DS-05: Model Registry (NEW)

- Table: Model Name, Version, Type (XGBoost/LightGBM), Status (Champion/Challenger/Archived), Deployed Date, Training Date, Training Rows, AUC-ROC
- **[Register New Model]** workflow: upload model artifact → validate → deploy as challenger
- Model lineage: version tree showing training data, hyperparameters, features used

---

## 7. Screens — Operations (Enhanced)

#### FR-V2-OPS-01: ETL Run History (Enhanced)

- Table: Run ID, Batch ID, Duration, Rows Received/Valid/Loaded/Rejected, Quality Score, Status
- Status icons: Completed ✓, Failed X, Running ⚙
- Click row → detailed audit record with step-by-step timing breakdown
- **NEW: Run comparison** — select two runs and diff the results

#### FR-V2-OPS-02: Data Quality Dashboard (Enhanced)

- Time-series chart: quality\_score over recent runs
- Threshold line at 90% (configurable)
- Breakdown: rejection reasons by run (stacked bar chart)
- **NEW: Schema drift alerts** — new/removed columns detected in source data

#### FR-V2-OPS-03: System Health (Enhanced)

- Service status cards: PostgreSQL, Redis, Feature Service (8002), State Service (8003), Prediction Service (8004), Decision Intel (8005), API Gateway (8080)
- Per service: Uptime %, last heartbeat, response time (ms), error rate (last hour)
- Green/Amber/Red status
- **NEW: Dependency graph** — visual showing which services depend on which

#### FR-V2-OPS-04: Full Audit Trail (NEW)

- Searchable, filterable, sortable table of all auditable actions
- Columns: Timestamp, User ID, Role, Action Type, Customer ID, Details (JSON expandable), IP Address
- Filter by: date range, user, role, action type, customer
- Export to CSV/PDF
- Retention policy indicator: "Records available from [date] to [date]"

---

## 8. Common UI Components (v2.0 Enhanced)

FR-V2-COM-01: Navigation

- Persistent left sidebar with role-based menu items:
  - **RM:** Dashboard, Priority Queue, My Customers, Portfolio
  - **Branch Manager:** Dashboard, Team Performance, Campaigns, Portfolio Analytics
  - **Data Scientist:** Model Ops, Feature Drift, Prediction Logs, Model Registry
  - **Operations:** ETL Pipeline, Data Quality, System Health, Audit Trail
  - **Admin:** User Management, API Keys, Sessions, Audit Trail
- Breadcrumb trail on every screen
- User avatar (initials) + name + role in top-right corner with logout dropdown

FR-V2-COM-02: Global Search

- Search bar (Ctrl+K shortcut): customer name, account number, customer ID
- Results dropdown: name, state icon, health score, branch, priority score
- Selecting → Customer Detail

FR-V2-COM-03: Notifications (Enhanced)

- Bell icon with unread count badge
- Dropdown: recent alerts with severity badge
- **NEW: WebSocket real-time** — alerts appear without page refresh
- Click → navigate to relevant screen
- "Mark all read" button

FR-V2-COM-04: Export

- Any table → CSV export
- Customer Detail → **PDF export** (one-page RM meeting summary) — PROMOTED
- PDF includes: customer profile, health score gauge, top NBA, SHAP summary, action history (last 5)

FR-V2-COM-05: Audit Logging (Enhanced)

- Every frontend action that modifies data writes to audit log
- Fields: timestamp, user ID, role, action type, customer ID, details, IP, session ID
- **NEW: Frontend action categories:** VIEW, CREATE, UPDATE, DELETE, EXPORT, LOGIN, LOGOUT

FR-V2-COM-06: Dark Mode (NEW)

- Toggle in user menu: Light / Dark / System
- Persisted to localStorage
- All charts, gauges, tables respect theme

---

9. API Contract — Frontend to Backend (v2.0)

9.1 Authentication

| #  | Endpoint      | Method | Role          | Purpose                          |
|----|---------------|--------|---------------|----------------------------------|
| A1 | /auth/login   | POST   | Public        | LDAP login → JWT + refresh token |
| A2 | /auth/refresh | POST   | Authenticated | Rotate access token              |

| #  | Endpoint                     | Method | Role          | Purpose                         |
|----|------------------------------|--------|---------------|---------------------------------|
| A3 | <a href="#">/auth/logout</a> | POST   | Authenticated | Blacklist token, revoke refresh |
| A4 | <a href="#">/auth/me</a>     | GET    | Authenticated | Current user profile + roles    |

## 9.2 Customers & Features

| #  | Endpoint                                              | Method | Role       | Purpose                                                                       |
|----|-------------------------------------------------------|--------|------------|-------------------------------------------------------------------------------|
| C1 | <a href="#">/api/customers</a>                        | GET    | RM, BM     | List RM's customers with latest features, state, health score, priority score |
| C2 | <a href="#">/api/customers/{customer_id}</a>          | GET    | RM, BM, DS | Full customer 360: profile + latest features + state + predictions + NBA      |
| C3 | <a href="#">/api/customers/{customer_id}/history</a>  | GET    | RM, BM, DS | Feature snapshots over time (date range)                                      |
| C4 | <a href="#">/api/customers/{customer_id}/timeline</a> | GET    | RM, BM, DS | State transition timeline with trigger reasons                                |
| C5 | <a href="#">/api/customers/{customer_id}/journey</a>  | GET    | RM, BM     | Customer journey: touchpoints, channels, product events                       |

## 9.3 Predictions & Explainability

| #  | Endpoint                                                         | Method | Role       | Purpose                                                       |
|----|------------------------------------------------------------------|--------|------------|---------------------------------------------------------------|
| P1 | <a href="#">/api/customers/{customer_id}/predictions</a>         | GET    | RM, BM, DS | Latest churn prob, CLV, health score with component breakdown |
| P2 | <a href="#">/api/customers/{customer_id}/predictions/history</a> | GET    | DS         | Prediction history over time for a customer                   |
| P3 | <a href="#">/api/customers/{customer_id}/shap</a>                | GET    | RM, BM, DS | SHAP waterfall values + LLM explanation for latest prediction |
| P4 | <a href="#">/api/customers/{customer_id}/markov/predict</a>      | GET    | RM, BM, DS | Markov next-state prediction with probabilities               |

## 9.4 Decision Intelligence (NBA)

| #  | Endpoint                                                      | Method | Role   | Purpose                                                   |
|----|---------------------------------------------------------------|--------|--------|-----------------------------------------------------------|
| D1 | <a href="#">/api/customers/{customer_id}/nba</a>              | GET    | RM, BM | Ranked NBA recommendations with LLM explanations          |
| D2 | <a href="#">/api/customers/{customer_id}/nba/alternatives</a> | GET    | RM, BM | Alternative actions (ranked 2-4 below top recommendation) |

| #  | Endpoint                              | Method | Role      | Purpose                                                                            |
|----|---------------------------------------|--------|-----------|------------------------------------------------------------------------------------|
| D3 | /api/customers/{customer_id}/priority | GET    | RM,<br>BM | Priority score breakdown:<br>Value × Urgency ×<br>Acceptance × Retention<br>Impact |

## 9.5 Actions & Feedback Loop

| #  | Endpoint                                                 | Method | Role      | Purpose                                                                                |
|----|----------------------------------------------------------|--------|-----------|----------------------------------------------------------------------------------------|
| F1 | /api/customers/{customer_id}/actions                     | GET    | RM,<br>BM | Action history<br>(paginated,<br>filterable by<br>type/date/outcome)                   |
| F2 | /api/customers/{customer_id}/actions                     | POST   | RM        | Log new action<br>taken by RM                                                          |
| F3 | /api/customers/{customer_id}/actions/{action_id}/outcome | PATCH  | RM        | Update action<br>outcome<br>(Contacted / Not<br>Reachable /<br>Accepted /<br>Declined) |
| F4 | /api/rm/{rm_id}/actions/summary                          | GET    | RM,<br>BM | Action summary<br>stats: total,<br>completed,<br>pending,<br>completion rate           |

## 9.6 Priority Queue

| #  | Endpoint                    | Method | Role | Purpose                                                           |
|----|-----------------------------|--------|------|-------------------------------------------------------------------|
| Q1 | /api/rm/{rm_id}/queue       | GET    | RM   | Daily priority queue: top N customers ranked by<br>Priority Score |
| Q2 | /api/rm/{rm_id}/queue/stats | GET    | RM   | Queue stats: critical/high/medium/low counts                      |

## 9.7 Alerts & Notifications

| #  | Endpoint                           | Method | Role          | Purpose                                                                       |
|----|------------------------------------|--------|---------------|-------------------------------------------------------------------------------|
| N1 | /api/alerts                        | GET    | RM, BM        | Unacknowledged alerts for<br>current user                                     |
| N2 | /api/alerts/{alert_id}/acknowledge | PATCH  | RM, BM        | Acknowledge an alert (timestamp<br>+ user ID)                                 |
| N3 | /ws/notifications                  | WS     | Authenticated | Real-time notification stream<br>(state changes, health drops, NBA<br>expiry) |

9.8 Branch Manager

| #  | Endpoint                                 | Method | Role | Purpose                                              |
|----|------------------------------------------|--------|------|------------------------------------------------------|
| B1 | /api/branch/{branch_code}/portfolio      | GET    | BM   | Branch-level portfolio aggregation                   |
| B2 | /api/branch/{branch_code}/kpis           | GET    | BM   | Branch KPIs with month-over-month deltas             |
| B3 | /api/branch/{branch_code}/team           | GET    | BM   | RM team performance table                            |
| B4 | /api/branch/{branch_code}/churn-forecast | GET    | BM   | Projected churn: 30/60/90 days with confidence bands |
| B5 | /api/branch/{branch_code}/heatmap        | GET    | BM   | Churn risk heatmap data (customers × weeks)          |

9.9 Campaigns

| #  | Endpoint                             | Method | Role   | Purpose                                                               |
|----|--------------------------------------|--------|--------|-----------------------------------------------------------------------|
| G1 | /api/campaigns                       | GET    | BM, DS | List active campaigns with performance metrics                        |
| G2 | /api/campaigns/{campaign_id}         | GET    | BM, DS | Campaign detail: response timeline, segment breakdown, RM performance |
| G3 | /api/campaigns/{campaign_id}/compare | GET    | BM, DS | Side-by-side campaign comparison data                                 |

9.10 Model Operations (Data Scientist)

| #  | Endpoint                                | Method | Role | Purpose                                                        |
|----|-----------------------------------------|--------|------|----------------------------------------------------------------|
| M1 | /api/models/performance                 | GET    | DS   | Champion model metrics (AUC-ROC, Precision, Recall, F1)        |
| M2 | /api/models/performance/trend           | GET    | DS   | Metric time-series: last 30/60/90 days                         |
| M3 | /api/models/performance/by-segment      | GET    | DS   | Model metrics broken down by customer segment                  |
| M4 | /api/models/champion-challenger         | GET    | DS   | Side-by-side comparison: traffic split, win rate, significance |
| M5 | /api/models/registry                    | GET    | DS   | Model registry: all versions, statuses, dates, metrics         |
| M6 | /api/models/registry                    | POST   | DS   | Register new model version                                     |
| M7 | /api/models/registry/{model_id}/promote | POST   | DS   | Promote challenger → champion                                  |

9.11 Feature Drift

| #  | Endpoint                               | Method | Role | Purpose                                                |
|----|----------------------------------------|--------|------|--------------------------------------------------------|
| R1 | /api/models/drift                      | GET    | DS   | Feature drift table: PSI scores, distributions, status |
| R2 | /api/models/drift/{feature_name}/trend | GET    | DS   | Single feature PSI trend over time                     |

## 9.12 Prediction Logs

| #  | Endpoint                     | Method | Role | Purpose                                                   |
|----|------------------------------|--------|------|-----------------------------------------------------------|
| L1 | /api/predictions/logs        | GET    | DS   | Searchable prediction log browser (paginated, filterable) |
| L2 | /api/predictions/logs/export | GET    | DS   | Export prediction logs as CSV                             |

## 9.13 ETL Operations

| #  | Endpoint                    | Method | Role | Purpose                                              |
|----|-----------------------------|--------|------|------------------------------------------------------|
| E1 | /api/etl/runs               | GET    | Ops  | Paginated ETL run history with dashboard KPIs        |
| E2 | /api/etl/runs/{run_id}      | GET    | Ops  | Detailed run audit: step timings, row counts, errors |
| E3 | /api/etl/runs/compare       | GET    | Ops  | Diff two ETL runs                                    |
| E4 | /api/etl/quality/trend      | GET    | Ops  | Data quality score time-series                       |
| E5 | /api/etl/quality/rejections | GET    | Ops  | Rejection reason breakdown by run                    |

## 9.14 System Health

| #  | Endpoint                          | Method | Role       | Purpose                                                            |
|----|-----------------------------------|--------|------------|--------------------------------------------------------------------|
| H1 | /api/system/health                | GET    | Ops, Admin | All service statuses: uptime, heartbeat, response time, error rate |
| H2 | /api/system/health/{service_name} | GET    | Ops, Admin | Single service detailed health                                     |

## 9.15 Audit Trail

| #  | Endpoint               | Method | Role       | Purpose                                              |
|----|------------------------|--------|------------|------------------------------------------------------|
| U1 | /api/audit/logs        | GET    | Admin, Ops | Full audit trail (paginated, filterable, searchable) |
| U2 | /api/audit/logs/export | GET    | Admin, Ops | Export audit logs as CSV                             |

## 9.16 Admin

| #  | Endpoint                     | Method   | Role  | Purpose             |
|----|------------------------------|----------|-------|---------------------|
| X1 | /admin/users                 | GET/POST | Admin | List / create users |
| X2 | /admin/users/{user_id}/roles | PATCH    | Admin | Update user roles   |

| #  | Endpoint                        | Method   | Role  | Purpose                                            |
|----|---------------------------------|----------|-------|----------------------------------------------------|
| X3 | /admin/api-keys                 | GET/POST | Admin | List / create API keys for service-to-service auth |
| X4 | /admin/api-keys/{key_id}/revoke | POST     | Admin | Revoke an API key                                  |
| X5 | /admin/sessions                 | GET      | Admin | Active user sessions                               |

## 10. WebSocket Protocol (NEW in v2.0)

### 10.1 Connection

```
ws://{host}:8080/ws/notifications
Headers: Authorization: Bearer {jwt}
```

### 10.2 Server → Client Messages

```
{
  "type": "alert",
  "severity": "critical|high|info",
  "title": "Health Score Drop",
  "message": "Customer John Doe health score dropped from 72 to 48",
  "customer_id": "CUST00123",
  "timestamp": "2026-08-06T14:30:00Z",
  "alert_id": "ALT-0042"
}
```

### 10.3 Message Types

| Type           | Trigger                                         | Navigation Target         |
|----------------|-------------------------------------------------|---------------------------|
| state_change   | Customer transitions to At Risk/Dormant/Churned | Customer Detail           |
| health_drop    | Health Score drops >20 points in one period     | Customer Detail           |
| nba_expiry     | NBA recommendation expires without action       | Customer Detail → NBA tab |
| batch_complete | Nightly batch prediction completes              | Dashboard                 |
| model_drift    | Feature drift exceeds threshold (DS only)       | Feature Drift Monitor     |
| etl_failure    | ETL pipeline run fails (Ops only)               | ETL Run History           |

## 11. v2.0 Scope Boundaries

### Included in v2.0

- ALL v1.0 PoC features (already built)
- Branch Manager Dashboard (promoted from deferred)
- Data Scientist: Full Model Ops (promoted from deferred)

- SHAP waterfall + LLM explanations
- Priority Queue with Priority Score
- Campaign Management basic views
- Customer Journey visualizer (Sankey/flow)
- Action outcome tracking (feedback loop)
- PDF export for customer summary
- WebSocket real-time notifications
- Full audit trail viewer
- Dark mode
- All 74 API endpoints documented above

Deferred to v2.1+

- Reinforcement Learning feedback UI (adaptive action selection)
- Multi-armed bandit experiment configuration UI
- Real-time streaming dashboard (Kafka-backed)
- Mobile/responsive layout
- RBAC fine-grained permission editor UI
- Custom dashboard builder (drag-and-drop widgets)
- Voice-to-text action logging
- Offline/PWA full support

12. Frontend State Management (Pinia Stores — v2.0)

| Store              | Responsibility            | Key State                                                  |
|--------------------|---------------------------|------------------------------------------------------------|
| useAuthStore       | Authentication + session  | user, roles, token, refreshToken, isAuthenticated          |
| useCustomerStore   | Customer data             | customers, selectedCustomer, filters, pagination           |
| usePredictionStore | Predictions + SHAP        | predictions, shapValues, llmExplanation, healthScore       |
| useNBASore         | NBA recommendations       | recommendations, alternatives, priorityScores              |
| useActionStore     | Action logging + outcomes | actions, outcomes, actionStats                             |
| useQueueStore      | Priority queue            | queue, queueStats                                          |
| useAlertStore      | Alerts + WebSocket        | alerts, unreadCount, wsConnection                          |
| useBranchStore     | Branch manager data       | branchKPIs, teamPerformance, churnForecast, heatmap        |
| useModelStore      | Model operations          | modelMetrics, championChallenger, modelRegistry, driftData |
| useCampaignStore   | Campaign management       | campaigns, campaignDetail                                  |
| useETLStore        | ETL operations            | etlRuns, qualityTrend, rejectionBreakdown                  |
| useSystemStore     | System health             | serviceStatuses, dependencyGraph                           |
| useAuditStore      | Audit trail               | auditLogs, filters                                         |
| useUIStore         | UI preferences            | theme (light/dark), sidebarCollapsed, notifications        |

### 13. API Response Envelope (Standard)

All v2.0 API responses follow a consistent envelope:

```
{
  "data": { ... },
  "meta": {
    "page": 1,
    "limit": 25,
    "total": 249,
    "pages": 10
  },
  "errors": null,
  "timestamp": "2026-08-06T14:30:00Z"
}
```

Error responses:

```
{
  "data": null,
  "meta": null,
  "errors": [
    {
      "code": "CUSTOMER_NOT_FOUND",
      "detail": "No customer with ID 'CUST99999'",
      "path": "/api/customers/CUST99999"
    }
  ],
  "timestamp": "2026-08-06T14:30:00Z"
}
```

### Appendix A: Endpoint Summary Matrix

| Category                     | Endpoint Count | New in v2.0                |
|------------------------------|----------------|----------------------------|
| Authentication               | 4              | 1 (/auth/me)               |
| Customers & Features         | 5              | 2 (journey, timeline)      |
| Predictions & Explainability | 4              | 3 (SHAP, Markov, history)  |
| Decision Intelligence (NBA)  | 3              | 2 (alternatives, priority) |
| Actions & Feedback           | 4              | 3 (outcome, summary)       |
| Priority Queue               | 2              | 2                          |
| Alerts & Notifications       | 3              | 1 (WebSocket)              |
| Branch Manager               | 5              | 5 (all promoted)           |
| Campaigns                    | 3              | 3                          |
| Model Operations             | 7              | 7 (all promoted + new)     |

| Category        | Endpoint Count | New in v2.0                     |
|-----------------|----------------|---------------------------------|
| Feature Drift   | 2              | 2                               |
| Prediction Logs | 2              | 2                               |
| ETL Operations  | 5              | 3 (detail, compare, rejections) |
| System Health   | 2              | 2                               |
| Audit Trail     | 2              | 2                               |
| Admin           | 5              | 5                               |
| TOTAL           | 58             | 37 new/changed                  |

## Appendix B: Backend Service Mapping

| Frontend Domain                   | Backend Service(s)                          | Port | Status       |
|-----------------------------------|---------------------------------------------|------|--------------|
| Auth                              | API Gateway (auth routes)                   | 8080 | ✔ Live       |
| Customers / Features              | Feature Engineering Service                 | 8002 | ✔ Live       |
| State / Timeline / Markov         | Customer State Service (Layer 1)            | 8003 | ✔ Live       |
| Predictions / SHAP / Health Score | Prediction Service (Layer 2)                | 8004 | 🔧 In Build   |
| NBA / Priority / LLM Explanations | Decision Intelligence Service (Layer 3)     | 8005 | 🔧 Scaffolded |
| Campaigns                         | Campaign Engine (Decision Intel sub-module) | 8005 | 🔧 Scaffolded |
| ETL / Data Quality                | ETL Engine                                  | —    | ✔ Live       |
| Model Registry / Drift            | Model Management Service                    | —    | 🔧 In Build   |
| Audit Trail                       | Gateway (middleware) + ETL audit tables     | 8080 | ✔ Live       |
| WebSocket Notifications           | Gateway (WS endpoint) + Redis pub/sub       | 8080 | 🔧 To Build   |
| System Health                     | Gateway health + per-service /health        | 8080 | ✔ Live       |